ADMISSION PROCESS IN HEALTHCARE

A PROJECT REPORT

Submitted by

KARTHICK S (2303737720521033)

in partial fulfillment of the requirement

for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

K.S.RANGASAMY COLLEGE OF TECHNOLOGY

(An Autonomous Institution, affiliated to AnnaUniversity Chennaiand Approved by AICTE, NewDelhi)

TIRUCHENGODE–637215

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BONAFIDE CERTIFICATE

Certified that this project report titled **"ADMISSION PROCESS IN HEALTHCARE"**isthebonafideworkof **KARTHICK S (2303737720521033)**who carried out the project under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertationonthebasisofwhichadegree or award was conferred on an earlier occasion on this or any other candidate.

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I declare that the project report on **"ADMISSION PROCESS IN HEALTHCARE"** is the result of original work done by me and best of my knowledge. This project report is submitted on the partial fulfillment of the requirement of the award of Degree of **B.Tech** in **Information Technology.**

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Place:Tiruchengode

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ABSTRACT

In the modern healthcare environment, efficiency, accuracy, and speed are critical when it comes to patient data management. One of the most fundamental and repetitive tasks in hospitals and clinics is the **patient admission process**, which typically involves gathering patient details, verifying them, storing them securely, and sending confirmation notifications. Traditionally, these processes are handled manually, consuming significant time and resources, while also being prone to human error. This project introduces an automated solution using **UiPath**, a leading Robotic Process Automation (RPA) platform, to handle the admission process seamlessly. The automation work flow is designed to read patient details including name, email, and ID from an Excel spreadsheet. It processes and validates the data, and then sends a personalized confirmation email to each patient using the SMTP email protocol. The final validated data is stored in a separate Excel file, ensuring organized and error-free record-keeping. The project also includes error handling and logging mechanisms to monitor process flow and capture exceptions, making the system reliable and robust. With this automation, healthcare providers can minimize manual involvement in repetitive tasks, reducethe riskofdata entry errors, andprovidequickerresponses topatients. This mini project demonstrates the potential of RPA in transforming routine administrative operations in healthcare. It highlights how tools like UiPath can be used not just to save time, but also to improve overall data integrity, compliance, and patient satisfaction. As digital transformation becomes increasingly important, such automated solutions can play a vital role in modernizing the healthcare industry. This system is scalable and can be integrated with other hospital management modules. It also foundation for future enhancements like AI-based data analysis and chatbot integration.

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CHAPTER 1

INTRODUCTION

Intoday'sfast-pacedanddigitallydrivenworld,thehealthcareindustryisundergoingasignificant transformation to improve patient care and operational efficiency. One of the most crucial components in any healthcare facility is the **admission process**, which involves collecting, verifying, and storing essential patient information. Traditionally, this process has been manual, time-consuming,andpronetohumanerror.Withthegrowingdemandforaccuracyandefficiency, **RoboticProcessAutomation(RPA)**offersapowerfulsolutiontostreamlineandautomatethese repetitive tasks. This project, titled **"Admission Process in Healthcare"**, aims to automate the patientadmissionworkflowusing**UiPath**,aleadingRPAtool.Theautomationprocessbeginsby readingpatient details suchas Name,Email,and Patient IDfrom an Excel file. These details are then verified, organized, and processed to ensure proper formatting and completeness. For valid entries, a personalized confirmation email is automatically sent to each patient, acknowledging their registration. Finally, the processed data is written back into another Excel sheet for record-keeping and future reference. The automation not only reduces the workload on administrative staff but also ensures **speed, accuracy, and reliability** in handling patient records. It eliminates theneedformanualdataentryandrepetitiveemaildrafting,allowinghospitalstafftofocusmore on patient care and critical tasks. This project demonstrates how RPA can be effectively used in the healthcare sector to digitize operations and enhance service delivery. Overall, the project showcasesthepotentialofUiPathinsolvingreal-worldproblemsandhighlightshowautomation can bring about meaningful improvements in one of the most vital industries — healthcare.

1.1 ABOUT THEPROJECT

This project focuses on automating the patient admission process in a healthcare setup using UiPath. It reads patient information such as name, email, and patient ID from an Excel sheet. Thedataisvalidatedandformattedtoensureaccuracyandcompleteness.Forvalidemail addresses,thesystemsendsconfirmationemailsautomaticallyviaSMTP. Theautomationlogs eachsteptohelptracktheprocessandidentifyanyerrors.Processedrecordsarewrittenback to an Excel workbook for record-keeping. The use of UiPath eliminates the need for manual dataentryandemailsending.Thisimprovestheoverallspeed,accuracy,andefficiencyofthe admissionprocess.Thesolutionisscalableandcanbeadaptedtodifferenthealthcare environments.ItdemonstrateshowRPAcanimproveadministrativeworkflowsinthe healthcare industry.

1.2 PROJECTSCOPE

- AutomatedDataHandling:Automatetheextractionandvalidationofpatientdetailsfrom Excel files.
- EmailNotification:Automaticallysendregistrationconfirmationemailstopatients using SMTP.
- ErrorLogging: Logall exceptionsandinvaliddataentriesforreviewanddebugging.
- DynamicRecordStorage:WritevalidatedpatientrecordsintoaseparateExcelfilefor documentation.
- Scalability:Designedtoaccommodatelargevolumesofpatientdatawithminimalhuman intervention.

1.3 UIPATHOBJECTIVES

ThemainobjectiveofthisprojectistoutilizeUiPathtoautomatethepatientadmissionprocess inthehealthcaredomain.Theautomationisdesignedtoextract,validate,andorganizepatient details such as name, email, and ID from an Excel file using DataTable operations.. It sends confirmationemailstopatientsautomaticallythroughtheSMTPprotocol,eliminatingtheneed formanualcommunication.ThefinalstructureddataiswrittenintoanewExcelfileforrecord-keeping.

1.1 AIMANDSCOPE

The aim of this project is to automate the patient admission process in the healthcare sector usingUiPath.TheautomationwillhelpinreadingpatientdetailsfromanExcelfile,validating the information, and sending confirmation emails automatically. This reduces manual effort, minimizes human error, and improves process efficiency.

Scope:Thescope encompasses thefollowingkeyareas:

- Automatereadingofpatientdetails suchasname,email,andIDfromExcel.
- Validateandstoreprocessedpatient informationinanewExcel file.
- Automaticallysendconfirmationemails toeachpatientusing SMTP.
- Implementlogginganderrorhandlingforprocesstrackinganddebugging.
- Reducehumaninvolvement inrepetitiveadministrativetaskstoimproveaccuracy.

CHAPTER 2

LITERATUREREVIEW

1. DelineatedAnalysisofUiPath-BasedAdmissionProcessinHealthcare

The admission process in healthcare institutions is a foundational procedure that ensures patients are formally registered, their medical information is documented, and appropriate departments are notified for treatment. Traditionally, this process involves several manual steps,including collecting personal information, verifying medical history,entering datainto hospitalsystems,andgeneratingpatientidentificationrecords.Whileessential,theseactivities aretime-consumingandoftenrepetitive,leadingtoinefficienciesandpotentialhumanerrors. With the increasing volume of patients—especially during periods like the COVID-19 pandemic—themanualapproachcanoverwhelmhospitalstaff,delaycare,andreducepatient satisfaction.Furthermore,thedependenceonphysicalformsandin-personregistrationcreates bottlenecks and risks data mismanagement or loss.In response to these challenges, Robotic Process Automation (RPA) offers a modern solution that enhances speed, accuracy, and consistencyinpatientadmission.Byautomatingdataentry,emailcommunication,andrecord generation, RPA tools like UiPath reduce the burden on administrative staff and streamline workflows.

2. RoboticProcessAutomation:AnOverviewandApplicationinPersonal Productivity – The Case of Admission Process in Healthcare

RoboticProcessAutomation(RPA)isatechnologythatenablestheautomationofrule-based, repetitivetasksby mimickinghuman interactions withdigitalsystems.RPA botscan operate acrossapplications,manipulatedata,triggerresponses,andcommunicate withothersystems. Its core advantage lies in improving efficiency, reducing human error, and freeing up professionals to focus on more strategic tasks. In the healthcare domain, especially in administrative processes like patient admission, RPA has proven to be a powerful tool. By automating the admission process, RPA significantly enhances personal productivity and institutional efficiency. Tasks such as reading patient details, validating entries, sending confirmationemails,and storingdata—allofwhicharepronetodelaysanderrorswhendone manually—can be handled automatically and consistently using platforms like UiPath.

3. AReviewonDataHandlingandAutomationforAdmissionProcess

The admission process in healthcare is a critical and complex task that involves handling a large volume of patient data, including personal details, medical history, insurance information,andconsentforms.Traditionally,thisprocesshasbeenmanual,time-consuming, and prone to human errors, leading to delays and inefficiencies. Efficient data handling is essential to ensure accurate patient records, compliance with healthcare regulations, and timely admission. Robotic Process Automation (RPA) offers a transformative solution by automating repetitive and rule-based tasks within the admission workflow. Using UiPath, healthcare institutions can streamline data entry, verification, and validation processes by automatically extracting data from various sources such as electronic forms, emails, and legacysystems. Automationreducesthedependencyonmanual effort, minimizeserrors,and accelerates patient onboarding. Moreover, RPA enhances data consistency and security by enforcingstandarddataformatsandenablingaudittrails. Insummary,theintegrationofRPA in healthcare admissions not only optimizes operational efficiency but also improves the overall patient experience through faster and more reliable data handling.

4.TemporalandFlexibleAutomationofAdmissionProcessUsingUiPath

The admission process in healthcare is highly dynamic, often requiring timely actions and adjustmentsbasedonpatientneeds andadministrativepriorities.Temporal automationrefers to the ability of the system to perform tasks based on specific time schedules or triggers, ensuring that critical steps in the admission workflow occur promptly without manual intervention. UiPath facilitates such temporal automation by allowing workflows to be scheduled or triggered by events, enabling the system to automatically process patient data, send notifications, and update records at the right moments. Flexible automation in the admission process means adapting to variations in data inputs, document formats, and exceptions without disrupting the workflow. UiPath’s capabilities, such as AI integration, cognitive document processing, and decision-making logic, enable the automation to handle diverse admission scenarios smoothly. For instance, if a patient’s documents are incomplete or require verification, the system can route the case for manual review or automatically request additional information, ensuring flexibility in handling exceptions.

5. Research on Software Testing Techniques and Automation Testing for Admission Process in Healthcare

Software testing is a crucial phase in the development and deployment of automation solutions for healthcare admission processes. It ensures that the automated workflows operate accurately, efficiently, and reliably, safeguarding sensitive patient data and maintaining compliance with healthcare standards. Various testing techniques, including functionaltesting,performancetesting,andsecuritytesting,playvitalrolesinvalidatingthe admission automation system. Functional testing verifies that each component of the admission process, such as data entry, validation, and notification modules, performs as expected. Performance testing assesses the system's response time and stability under varyingloads,importantinhandlingpeakadmissionperiods.Securitytestingensurespatient dataconfidentialityandcompliancewithregulationslikeHIPAAbydetectingvulnerabilities and preventing unauthorized access. Automation testing enhances the testing process by using tools to execute repetitive test cases faster and with higher accuracy than manual testing.Fortheadmissionprocess,UiPath'stestingframeworkcanbeemployedtosimulate real-worldadmissionscenarios,validatedatahandling,andidentifyworkflowerrorsearlyin the deployment cycle. Automation testing not only reduces testing time but also increases coverage, making the admission process more robust and reliable.

6. Recent Trends in Automation – A Study of Admission Process in Healthcare Development Using UiPath

Thehealthcaresectorisincreasinglyadoptingautomationtoenhanceoperationalefficiency, reduce errors, and improve patient experiences. Robotic Process Automation (RPA), particularly through platforms like UiPath, is at the forefront of this transformation, especiallyinpatientadmissionprocesses.Traditionalautomationrelieson predefinedrules, butrecentadvancementshaveintroducedAgenticAI,enablingsystemstomakeautonomous decisions. UiPath's integration of Agentic AI allows bots to handle complex tasks such as medicalpolicyqualityassurance,revenuecyclemanagement,andclaimsprocessingwithout human intervention. This shift empowers healthcare organizations to streamline operations and reduce manual workloads. Handling unstructured data, such as handwritten forms and medical records, has been a challenge in automation. UiPath's Document Understanding capabilities leverage AI to extract and interpret data from diverse document types.

CHAPTER 3

METHODOLOGYANDIMPLEMENTATION

3.1 EXISTINGSYSTEM

In the existing healthcare admission system, most tasks are performed manually by administrativestaff.Theprocessinvolvescollectingpatientinformationthroughpaperformsor manual dataentryinto electronicsystems. Thisincludes verifyingpersonal details, insurance information, medical history,and consent forms. Duetothemanual nature ofthesetasks,the process is often time-consuming, prone to human errors, and may lead to delays in patient admission. Moreover, data handling across multiple systems, such as hospital management software, billing, and insurance portals, requires repetitive efforts to ensure consistency and accuracy.Lackofintegrationbetweenthesesystemsfurthercomplicatestheworkflow,causing inefficienciesandsometimesresultinginduplicatedataentry. Themanualadmissionprocess also involves physical document management, which increases the risk of lost or misplaced patientrecords.Additionally,administrativestaffareburdenedwithrepetitiveandrule-based tasks, reducing their availability for more critical patient-centered activities.

3.2 PROPOSEDSYSTEM

TheproposedsystemleveragesRoboticProcessAutomation(RPA)usingUiPathtoautomate the entire admission process in healthcare. This automation aims to streamline data entry, verification,andmanagementtasksbyreducingmanualinterventionandimprovingaccuracy. UiPathrobotswillextractpatientinformationfromdigitalforms,emails,andlegacysystems, automatically validate the data, and update hospital databases in real time. By automating repetitive, rule-based tasks such as data capture, document verification, and record updating, the system minimizes human errors and speeds up patient admissions. The automation workflows are designed to be flexible, allowing the system to handle exceptions such as incomplete or incorrect data by routing such cases for manual review.

3.3 MODULES

- AutomationTools
- AutomationEngine
- UiPath

3.3.1 Automationtools

Automationtoolsaresoftwareapplicationsthathelpautomaterepetitiveandrule-basedtasks, improving efficiency and accuracy. In the healthcare admission process, these tools reduce manual data entry and minimize errors. UiPath is a leading RPA tool that provides an easy-to-use platform for designing, deploying, and managing automation workflows. It supports integration with multiple systems and offers features like AI and machine learning for intelligentautomation.Usingsuchtoolshelpshealthcareorganizationsstreamlineoperations and enhance patient service quality.

3.3.2 Automationengine

An automation engine is the core component that executes the automation workflows designed using RPA tools. In UiPath, the automation engine runs bots that interact with applications,processdata,andperformtasksasdefinedintheworkflow.Itensuresaccuracy, speed, and consistency in task execution without human intervention. This engine is crucial for handling large-scale automation in processes like patient admissions in healthcare.

3.4 OVERVIEWOFRPA

RoboticProcessAutomation(RPA)isatechnologythatusessoftwarebotstoautomaterepetitive, rule-based tasks without human involvement. It mimics human actions such as clicking, typing, and reading data from screens. RPA improves efficiency, reduces errors, and saves time across business processes. In healthcare, RPA can automate tasks like patient admission, billing, and data entry. Tools like UiPath make it easy to build and manage these automation workflows.

3.5 RPAFEATURES

Robotic Process Automation (RPA) offers a variety of powerful features that make it an ideal technologyforautomatingrepetitiveandrule-basedtasks,suchasthoseinvolvedinaBirthday Reminder Bot. Key features of RPA that support the development and operation of such a bot include:

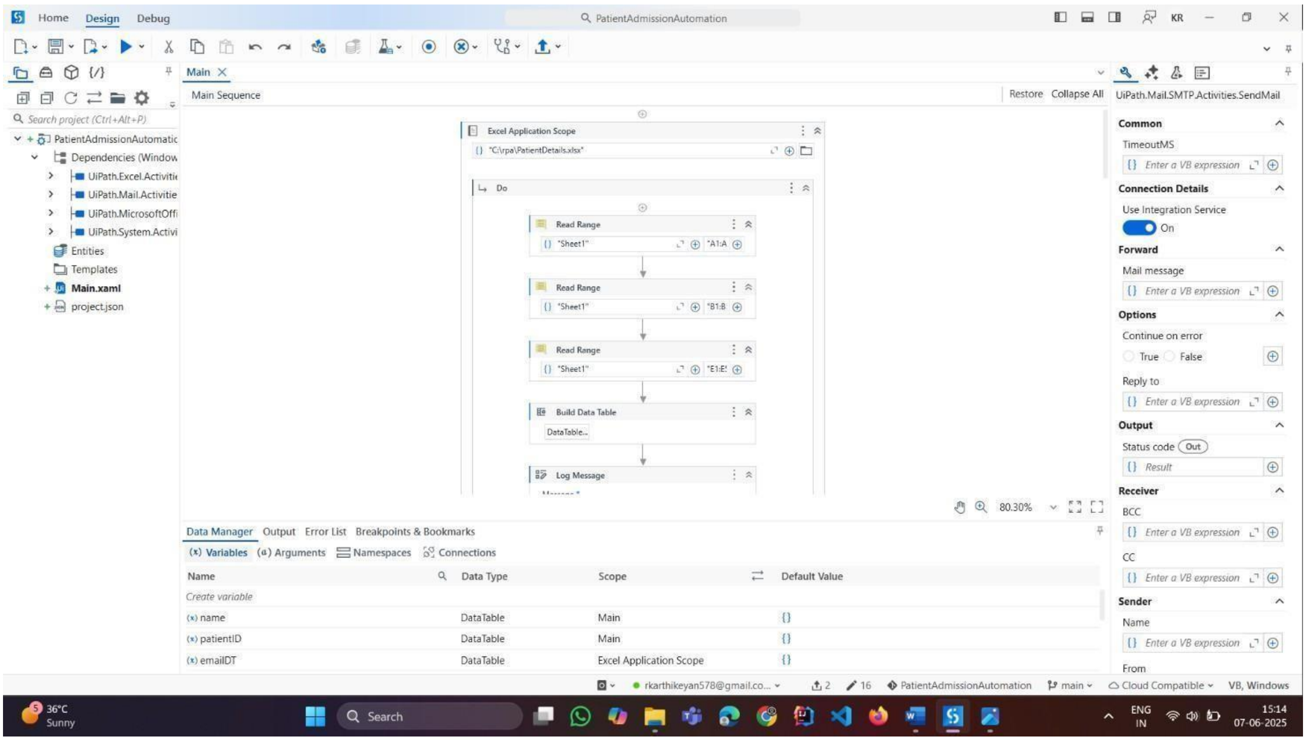
- **User-Friendly Interface:** RPA tools like UiPath offer intuitive, drag-and-drop interfaces that allow even non-programmers to design and automate workflows easily.

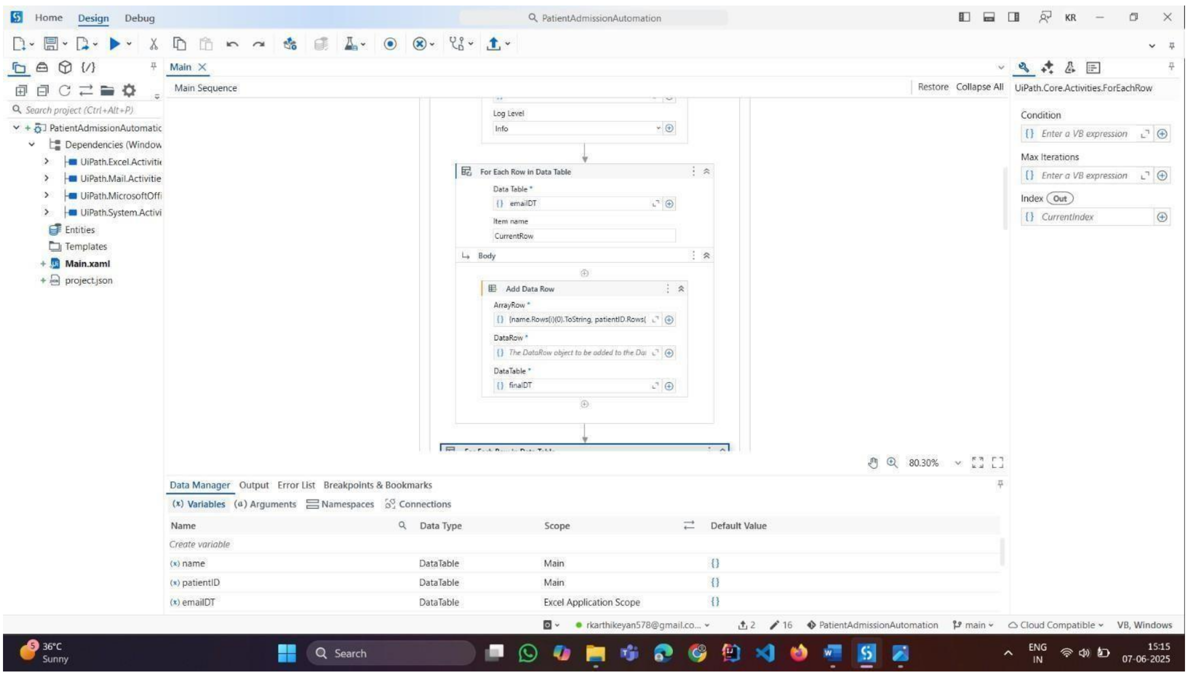
- **Bot Scalability:** RPA bots can be scaled up or down based on workload demands, allowing organizations to handle varying volumes of data and tasks efficiently.
- **Cross-Application Integration:** RPA can interact with multiple applications, including web, desktop, ERP systems, and legacy platforms, without the need for APIs or code changes.
- **Rule-Based Task Automation:** It excels at automating high-volume, rule-based taskssuchasdataentry,formfilling,anddatavalidation,reducingmanualworkload and human error.
- **Error Handling and Logging:** RPA tools provide built-in error handling mechanisms and detailed logging features, making it easy to monitor, debug, and improve processes.
- **SecurityandCompliance:**RPAensuresdataconfidentialitywithrole-basedaccess, audittrails,andsecurecredentialstorage,whichiscrucialindomainslikehealthcare.
- **AIandMachineLearningIntegration:**Modern RPAplatformsintegratewithAI and ML to handle semi-structured and unstructured data, enabling intelligent automation for decision-making processes.
- **Task Scheduling:**RPA tools allow automation tasks to be scheduled at specific timesortriggeredbycertainevents,ensuringtimelyexecutionwithoutmanualinput.
- **AnalyticsandReporting:**RPAplatformsprovidebuilt-inanalyticsdashboardsand reporting features to track bot performance, measure ROI, and identify areas for improvement.

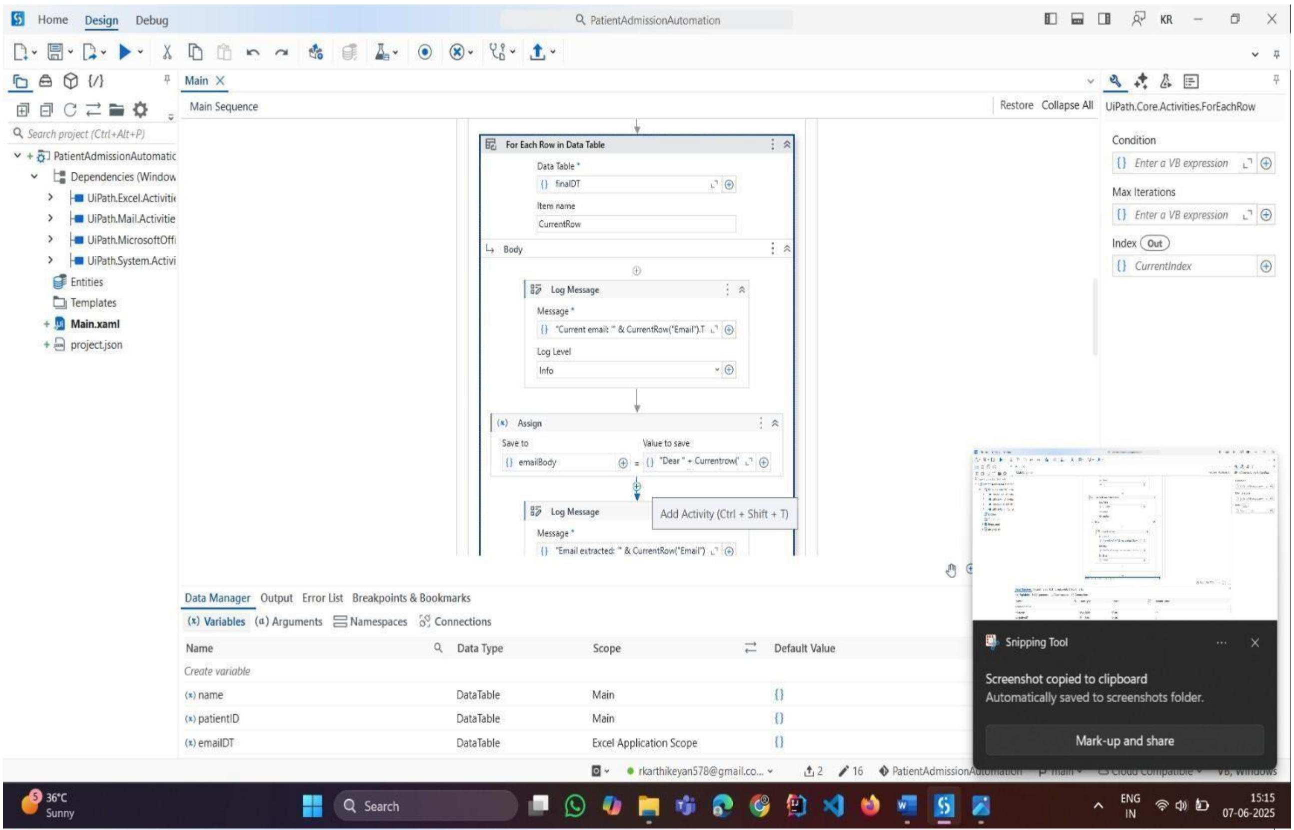
CHAPTER 4

RESULTANDDISCUSSION

4.1 RESULTS



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Debug
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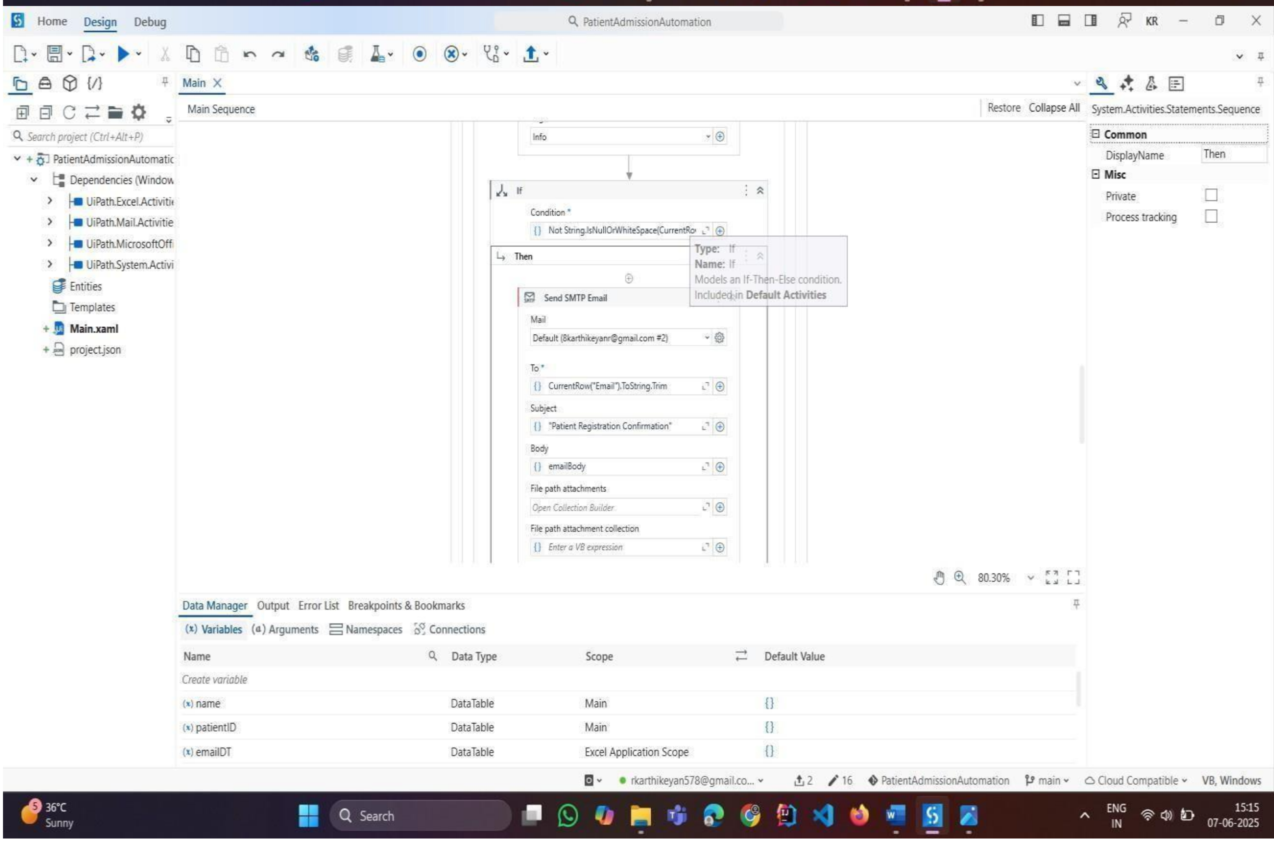
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Included in Default Activities
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Default (&karthikeyanr@gmail.com #2)
To *
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Subject
"Patient Registration Confirmation"
Body
emailBody
File path attachments
Open Collection Builder
File path attachment collection
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Output
Error List
Breakpoints & Bookmarks
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Arguments
Namespaces
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Name
Data Type
Scope
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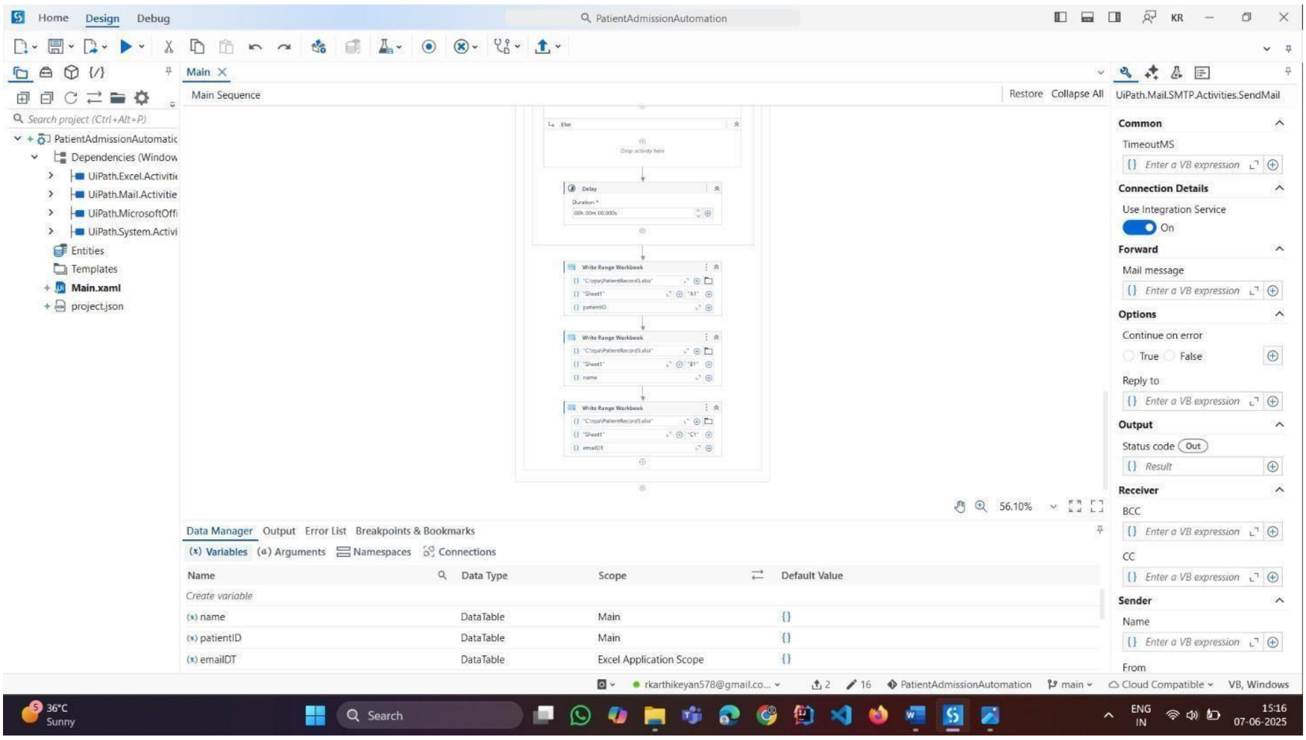
07-06-2025



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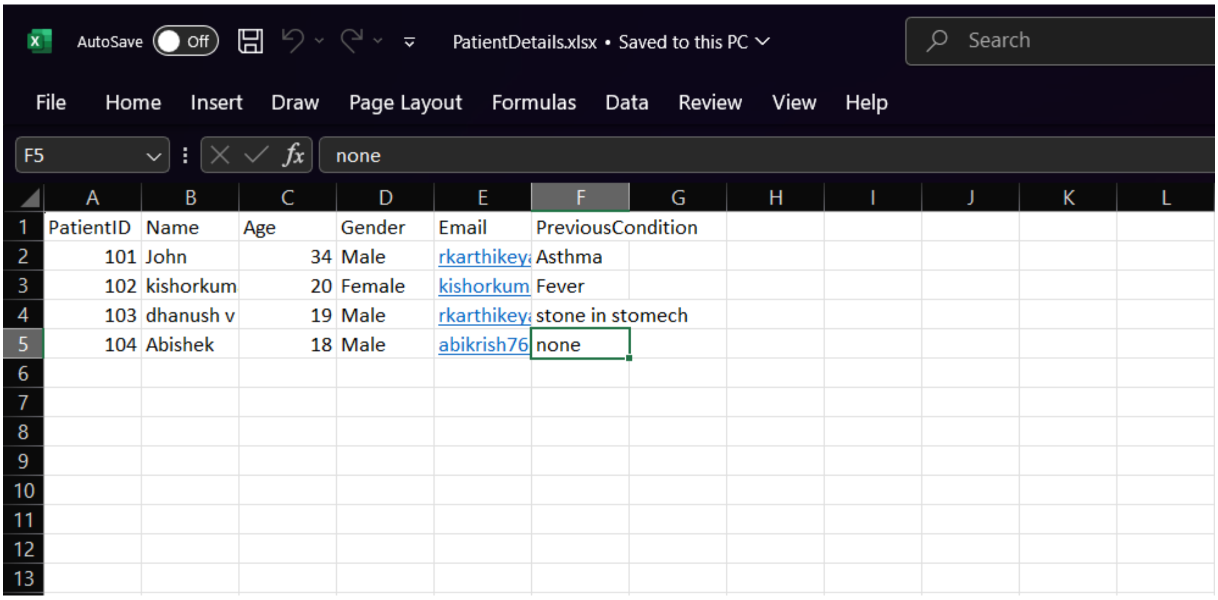

PatientID	Name	Age	Gender	Email	PreviousCondition
101	John	34	Male	rkarthikey	Asthma
102	kishorkum	20	Female	kishorkum	Fever
103	dhanush v	19	Male	rkarthikey	stone in stomech
104	Abishek	18	Male	abikrish76	none

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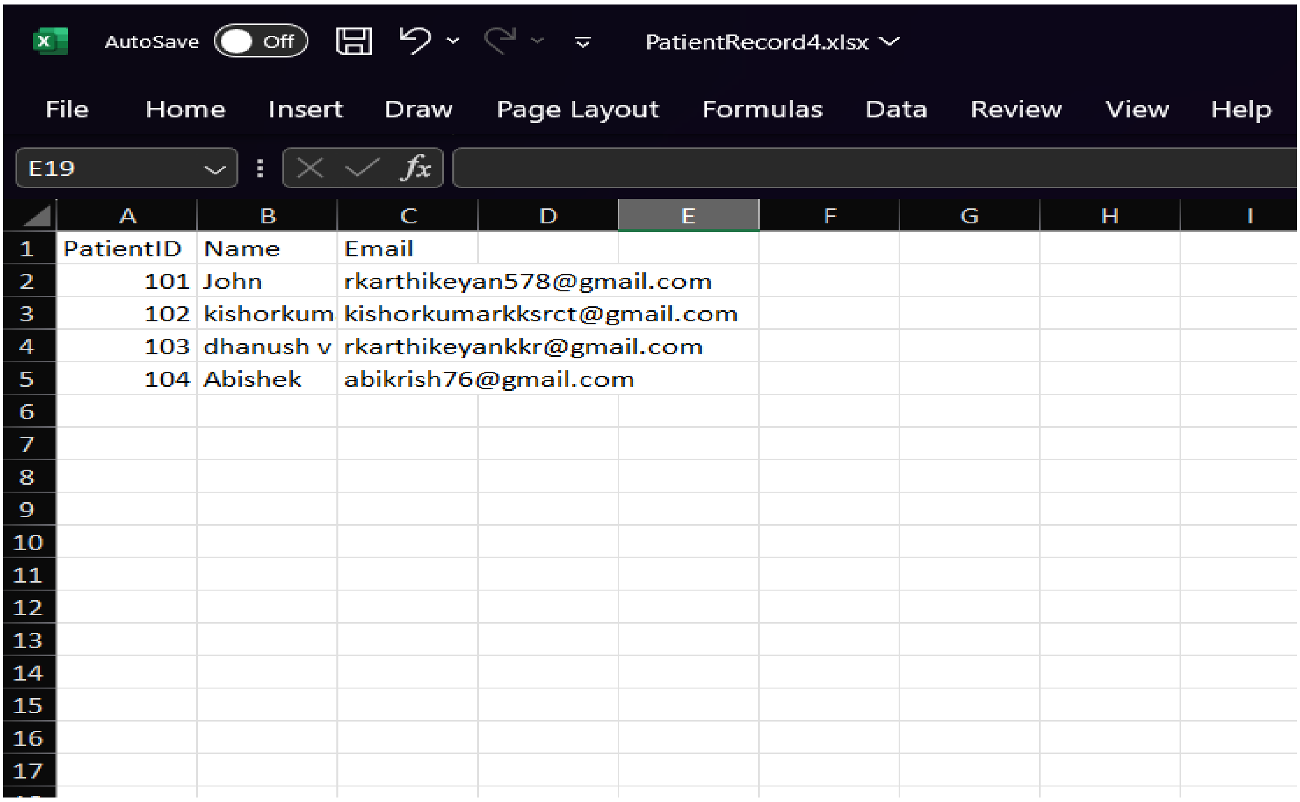
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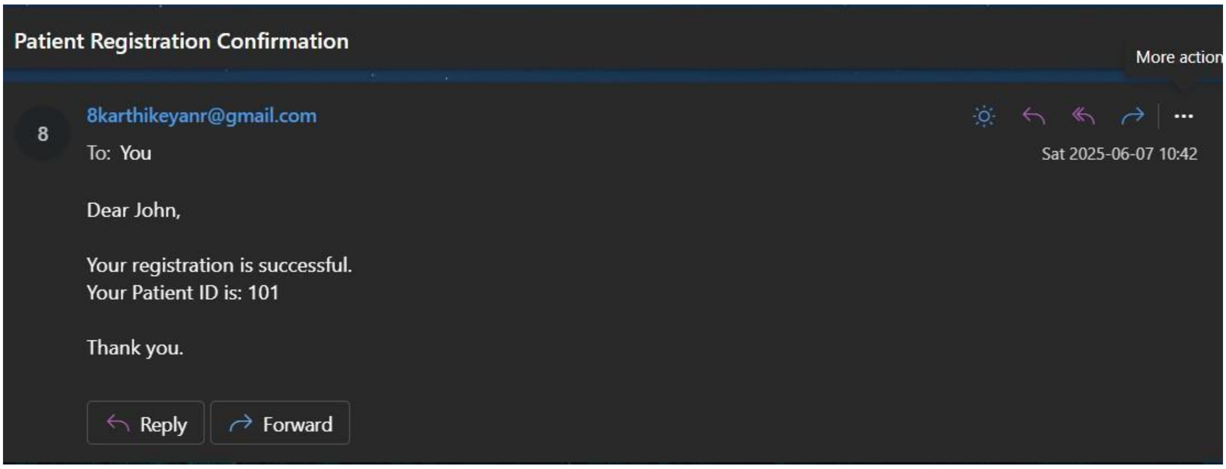
Patient Registration Confirmation
More action
8karthikeyanr@gmail.com
To: You
Sat 2025-06-07 10:42
Dear John,
Your registration is successful.
Your Patient ID is: 101
Thank you.
Reply
Forward

AFTEREXECUTION

4.2 DISCUSSION(ABOUTACTIVITYKEYS):

In mini project on automating the admission process in healthcare using UiPath, activity keys play a vital role in designing and executing the workflow. Activity keys are the core functions or components that enable the UiPath robot to perform tasks like collecting patient information, validating inputs, updating records, and sending confirmations. We used key activities such as **'Read Range'** to extract patient details from Excel sheets, and **'Input Dialog'** to collect real-time inputs like patient ID or department. The **'Assign'** activity was used to store and manipulate data variables,while**'If'**conditionscontrolledtheflowbasedoninsurancestatusormissinginformation. Forinteractingwithhospitalsystemsoronlineforms,activitieslike **'Click'**,**'TypeInto'**,and **'Get Text'** were essential. To ensure automation continuity and error handling, we implemented **'Try Catch'**, **'Message Box'**, and **'Log Message'** activities, which helped in debugging and tracking workflow execution. These activity keys provided structure, logic, and interaction capabilities throughout the admission process. These activity keys provided structure, logic, and interaction capabilities throughout the admission process. By combining these activities in sequences and flowcharts,weachievedasmooth,error-freeautomationprocess.Overall,activitykeyswerecrucial in transforming manual admission tasks into a reliable and efficient automated workflow in our healthcare scenario.

WORKFLOW:

1. StartMainSequence

➤Actsasthecentral containerfortheentireautomationlogicintheadmissionprocess.

➤Initializesrequiredvariables,setsuplogging,andhandlesoverallexecutionflow

2. OpenExcelApplicationScope

➤OpenstheExcelfilecontainingpatientadmissiondetailslikename,age,contact,and insurance status.

➤Ensuresallsessionsareproperlyclosedafterreadingorwritingtoavoidlocking issues.

3. ReadPatientDatafrom Excel

➤UsestheReadRangeactivityto importpatient admissionrecordsintoa DataTable.

➤Allowsdynamicreadingofrowsforprocessingeachpatient'sdetailsduring admission.

4. ValidateandProcessPatientRecords

➤IteratesthrougheachrowusingForEachRowtoextractindividualpatient information.

➤UsesIfconditions toverifynecessaryfieldslikeID,age,andinsuranceavailability.

➤Flagsincompleteorincorrect entriesformanualverification orsendsalerts.

5. UpdateRecordsandSend Confirmations

➤Foreachvalidentry,updatesthehospitalmanagementsystemorExcelfile accordingly.

➤Generatesadmissionconfirmation usingstringformatting.

➤SendsautomatedemailsorSMS topatientsusing**SendOutlookMailMessage**or **SMTP**activities.

6. LoggingandExceptionHandling

➤Maintains alogofallsuccessfullyprocessedandfailedrecordsusing**Log Message** and**WriteLine**.

➤Uses**TryCatch**blockstocaptureexceptionsandpreventworkflowfailure.

➤Createsareportorlogfilewiththestatusofeachadmissionprocessedfortransparency and auditing.

CHAPTER 5

CONCLUSION

The automation of the admission process in healthcare using UiPath has demonstrated significant improvements in operational efficiency, data accuracy, and overall patient experience. By replacing manual, repetitive tasks with software bots, hospitals can reduce processing time, eliminate human errors, and ensure that patient records are handled securely and consistently. This project has shown thatUiPathisapowerfulandflexibletoolcapableofintegratingwithexistingsystemsandmanaging largevolumesofdataeffectively.Through theuseofcoreUiPathactivitiessuchasReadRange, For EachRow,If,Assign,andSendMail,theentireadmissionprocess—fromdataretrievalandvalidation toconfirmationandlogging—wassuccessfullyautomated.Theimplementationofexceptionhandling and logging ensures the system remains robust and traceable. This not only saves time and labor but alsoallowshospitalstafftofocusmoreonpatientcareratherthanadministrativeduties.Inconclusion, robotic process automation presents a transformative approach for healthcare institutions seeking to modernize and optimize their workflows. Our project stands as a practical example of how UiPath can contribute to a more streamlined, accurate, and patient-focused admission system.

FUTUREWORK:

WhilethisprojectsuccessfullyautomatesthecoreadmissionprocessinhealthcareusingUiPath,there is considerable potential for future enhancements. One key area of development is the integration of the automation workflow with **Electronic Health Records (EHR)** systems and **Hospital Management Systems (HMS)** to ensure seamless end-to-end data synchronization. This would eliminate the need for external Excel files and enable real-time data exchange between departments. Finally, future versions could include **cloud deployment** and **mobile accessibility**, allowing healthcare professionals to monitor and interact with the automation process remotely, enhancing accessibility and control.

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